1 A short proof that $\mathbb{Q}[x]/(f_n)$ has no totally real subfields

Lemma 1.1. *If $f \in \mathbb{Q}[x]$ has a unique real root that then $\mathbb{Q}[x]/(f)$ has no nontrivial totally real subfield.*

Proof. Consider $\overline{\mathbb{Q}}$ an algebraic closure of $\mathbb{Q}$ as a subfield of $\mathbb{C}$, let $G_{\mathbb{Q}}$ be the total Galois group $\mathrm{Gal}\left(\overline{\mathbb{Q}}/\mathbb{Q}\right)$ and let $\sigma \in G_{\mathbb{Q}}$ be the element induced by the complex conjugation on $\mathbb{C}$. Assume towards contradiction that $\iota : K \hookrightarrow \mathbb{Q}[x]/(f)$ is an embedding of a totally real field $K \neq \mathbb{Q}$. Then the restriction induced restriction map

$$\iota^* : \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(f), \overline{\mathbb{Q}}\right) \to \mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$$

is a surjective G-equivariant map. The totally realsness of K implies that σ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$, and hence, the G-equivariance of ι^* implies that σ reduces to an action on each fiber of ι^*. However, the fact that f has a unique real root implies that it has an odd degree, and hence, each fiber of ι^* has an odd cardinality. In particular, since σ is an involution, it must have at least one fixed point in each fiber,

which contradicts the fact that f has a unique real root. $\square$

Corollary 1.2. *For each n, the quotient $\mathbb{Q}[x] / \left(x^n + x^{n-1} + \cdots + x - 1\right)$ has no totally real subfields other than $\mathbb{Q}$.*

1.1 Generalization:

I will try to generalize it to the same generalization as in [1, Appendix A]

Lemma 1.3. *Let $E \leq, F, K \leq \overline{E}$ be field extensions such that $E \leq F$ is finite. Assume that there exists a prime p such that $p \nmid [F : E]$ and such that $Aut\left(\overline{E}, K\right)$ is a pro-p group. Then for any totally K sub-extension $E \leq L \leq F$,*

$$[L : E]_s \leq \left|\mathrm{Hom}_E\left(F, K^{ins}\right)\right|$$

2 Complete Proof

The indices of f_n has a shift by one from the indices in the rest of the files in the repository

For each n denote

$$f_n = x^n - x^{n-1} - \ldots - 1$$

It has been shown that f_n is irreducible over $\mathbb{Q}$ (I will add a citation later). We will show that for each n, the Galois group of f_n is isomorphic to S_n.

Fix an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$, a root $\alpha \in \overline{Q}$ of f_n and for each prime p, fix an algebraic closure $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ and an embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}_p}$ extending the natural embedding $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$. Denote by G the absolute Galois group $\mathrm{Gal}\left(\overline{\mathbb{Q}}/\mathbb{Q}\right)$, by G_n the Galois group of f_n, and for each prime p, denote by $I(p) \subseteq G$ the absolute inertia group at p, i.e., the embedding into G of the inertia group $\mathrm{Gal}\left(\overline{\mathbb{Q}_p}/\mathbb{Q}_p^{\mathrm{un}}\right)$, where $\mathbb{Q}_p^{\mathrm{un}}$ is the maximal unramified extension of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}_p}$.

Using this notation, we can first adapt part of Martin's proof of [2, Theorem 2.5] to show the following lemma:

Lemma 2.1. *For each odd prime p, $I(p)$ acts on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$ either trivially or as a single transposition.*

Proof. postponed. $\square$

In light of this lemma it makes sense to guess that the G_n contains a transposition, since otherwise the discriminant of $\mathbb{Q}(\alpha)$ would be ramified only at 2. We can show this by analyzing the discriminant d_n of f_n for each n. However, before doing so, we will use the previous lemma to derive a lemma which, as we will explain afterwards, will make the analysis of the discriminant even more useful:

Lemma 2.2. *There is no subextension of $\mathbb{Q}(\alpha)/\mathbb{Q}$ that is ramified above any odd prime.*

Proof. Let $\iota : K \hookrightarrow \mathbb{Q}(\alpha)$ be a proper subextension and let p be an odd prime. It suffices to show that any embedding of K into $\overline{\mathbb{Q}_p}$ lands inside $\mathbb{Q}_p^{\mathrm{un}}$ for which it suffices to show that $I(p)$ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$. If $I(p)$ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$ then this is immediate. Assume otherwise. Then by the previous lemma, $I(p)$ acts as a single transposition on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$. Moerover, the precomposition map

$$\iota^* : \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right) \to \mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$$

is a surjective G-equivariant map. In particular it is

- Surjective with fibers of constant size $[\mathbb{Q}(\alpha) : K] \geq 2$;
- Compatible with the action of $I(p)$.

However, the only way a transposition can be compatible with such a map is if it swaps two elements inside a single fiber and thus acts trivially on the image. $\square$

The reason this lemma makes the analysis of the discriminant even more useful, is that using the analysis of the discriminant, and using the Minkowski bound, we can in fact show that $\mathbb{Q}(\alpha)$ has no proper subextensions at all, from which we will derive that G_n acts primitively on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$. This is in turn useful since a primitive permutation group that contains a transposition is, as we shall also show, the full symmetric group, and thus we will be able to conclude that $G_n \cong S_n$. Without further ado then, we bravely proceed to analyze the discriminant:

Lemma 2.3.

$$v_2(d_n) = \begin{cases} n-1 & n \text{ is odd} \\ 0 & n \text{ is even} \end{cases}$$

Proof. By [2]

$$d_n = \frac{(-1)^{\binom{n+1}{2}} \left((n+1)^{(n+1)} - 2^{n+1} n^n\right)}{(n-1)^2}$$

If n is even it is easy to see that both the denominator and numerator are odd, and hence $v_2(d_n) = 0$ as required. Assume now that n is odd. Then we can manipulate the expression above and get

$$d_n = (-1)^{\binom{n+1}{2}} 2^{n+1} \cdot \frac{\left(\frac{n+1}{2}\right)^n - n^n}{(n-1)^2}$$

and hence if we denote $m = v_2(n-1)$ we get that

$$v_2(d_n) = n + 1 - 2m + v_2\left(\left(\frac{n+1}{2}\right)^n - n^n\right)$$

Therefore, our claim is equivalent to showing that

$$n - 1 = n + 1 - 2m + v_2\left(\left(\frac{n+1}{2}\right)^n - n^n\right)$$

or equivalently, that

$$v_2\left(\left(\frac{n+1}{2}\right)^n - n^n\right) = 2m-2$$

Assume first that $m=1$, then

$$v_2\left(\frac{n+1}{2}\right) \geq 1$$

and hence

$$v_2\left(\left(\frac{n+1}{2}\right)^n\right) \geq n > 0 = v_2\left(n^n\right)$$

and hence

$$v_2\left(\left(\frac{n+1}{2}\right)^n - n^n\right) = 0 = 2m-2$$

Assume now that $m \geq 2$. In the rest of the proof we will use greek letters solely to denote different odd integer whose exact value is unknown and immaterial. As a first example for this notation, we denote

$$n-1 \equiv \epsilon 2^m \mod 2^{2m-1}$$

From this we may deduce that

$$
\begin{aligned}
&\left(\frac{n+1}{2}\right)^{n+1} - n^n \equiv \\
&\left(\frac{\epsilon 2^m + 2}{2}\right)^{n+1} - \left(\epsilon 2^m + 1\right)^n \equiv \\
&\left(\eta 2^{m-1} + 1\right)^{n+1} - \left(\epsilon 2^m + 1\right)^n \equiv \backslash 3m - 3 \geq 2m - 1 \\
&\binom{n+1}{2} \eta^2 2^{2m-2} + (n+1)\,\epsilon 2^{m-1} + 1 - n\epsilon 2^m - 1 \equiv \\
&\frac{(\epsilon 2^m + 2)\,(\epsilon 2^m + 1)}{2} \eta^2 2^{2m-2} + (\epsilon 2^m + 2)\,\epsilon 2^{m-1} - (\epsilon 2^m + 1)\,\epsilon 2^m \equiv \\
&\iota \eta^2 2^{2m-2} + \epsilon^2 2^{2m-1} + \epsilon 2^m - \epsilon^2 2^{2m} - \epsilon 2^m \equiv \\
&2^{2m-2}\left(\iota \eta^2 - 2\epsilon^2\right) \\
&2^{2m-2}\alpha \mod 2^{2m-1}
\end{aligned}
$$

and in particular its 2-adic valuation is $2m - 2$ as required. □

A first easy corollary of this lemma is the following promised statement:

Corollary 2.4. *G_n contains a transposition for each $n \geq 2$.*

Proof. This proof works only for even n or n that are sufficiently large. We could also cite [2, Theorem 2.5]. However, in his proof, he dismisses

pretty much all the content of the following proof as "easy to see", and I am not convinced yet.

By lemma 2.1, it suffices to show that there is an odd prime that ramifies in $\mathbb{Q}(\alpha)$, or equivalently, that divides the discriminant of $\mathbb{Q}(\alpha)$.

Assume first that n is even this. Then by the previous lemma, d_n is odd. However, since $1, \alpha, \ldots, \alpha^{n-1}$ spans a sublattice of the ring of integers of $\mathbb{Q}(\alpha)$, the discriminant of $\mathbb{Q}(\alpha)$ divides d_n and is thus odd as well. Therefore, there is an odd prime , p, that divides the discriminant of $\mathbb{Q}(\alpha)$.

Assume now that n is odd. By the same argument as before,

$$\operatorname{disc}(\mathbb{Q}(\alpha)) \,|d_n$$

and hence

$$v_2(\operatorname{disc}(\mathbb{Q}(\alpha))) \leq v_2(d_n) = n-1$$

Therefore, it suffices to show that

$$\operatorname{disc}(\mathbb{Q}(\alpha)) > 2^{n-1}$$

By [3] for any sufficiently large n

$$\operatorname{disc}(\mathbb{Q}(\alpha)) > 22^n$$

so this is clearly true. I have not went through [3] carefully enough yet to see if they detail how large n needs to be, so I am not sure how feasible it will be to check the remaining small values of n computationally.

□

As promised, we can also use the previous lemma, to show that $\mathbb{Q}(\alpha)$ has no proper subextensions at all. While for even values of n this is immediate, in order to derive it for odd values of n we will first need to prove a lemma and two corolarries:

Lemma 2.5. *Let $K/\mathbb{Q}$ be a field extension of degree at least 3. Then*

$$disc(K) \geq 2^{[K:\mathbb{Q}]}$$

Proof. Denote $r = [K : \mathbb{Q}]$ and $d_K = \operatorname{disc}(K)$. It suffices to show that

$$\sqrt{d_K} \geq 2^{r/2}$$

However, by the Minkowski bound

$$\sqrt{d_K} \geq \left(\frac{\pi}{4}\right)^{r/2} \cdot \frac{r^r}{r!}$$

and hence it suffices to show that

$$\left(\frac{\pi}{4}\right)^{r/2} \cdot \frac{r^r}{r!} \geq 2^{r/2}$$

or equivalently that

$$\frac{r^r}{r!} \geq \left(\frac{8}{\pi}\right)^{r/2}$$

which is true for $r = 3$ and follows for larger r by induction. $\square$

Corollary 2.6. *If* $[K : \mathbb{Q}] \geq 2$ *and* K *is ramified only at* 2 *then*

$$v_2\left(d_K\right) \geq [K : \mathbb{Q}]$$

Proof. If $r > 3$ it follows directly from the lemma. Assume that $r = 2$. Then $K = \mathbb{Q}\left(\sqrt{d}\right)$ for some integer d. Since K is ramified only at 2, $d = \pm 2^k$ for some integer k. Therefore, one of three cases occures:

- $K = \mathbb{Q}\left(\sqrt{2}\right)$, in which case $d_K = 8$;

- $K = \mathbb{Q}\left(\sqrt{-2}\right)$, in which case $d_K = 8$;
- $K = \mathbb{Q}\left(\sqrt{-1}\right)$, in which case $d_K = 4$;

In each case $v_2\left(d_K\right) \geq 2$. $\square$

Even without analysing all the possible discriminants of quadratic extensions of $\mathbb{Q}$, we could use the previous lemma to show that $v_2\left(d_K\right) \geq [k : \mathbb{Q}] - 1$, and while the above corollary might seem only a slight improvement of this slightly weaker bound, the following corollary makes a very significant use of it:

Corollary 2.7. *There are no proper subextensions of* $\mathbb{Q}\left(\alpha\right)/\mathbb{Q}$.

Proof. Let $K \hookrightarrow \mathbb{Q}\left(\alpha\right)$ be a proper subextension. By Lemma 2.2, $K/\mathbb{Q}$ is ramified only at 2. Therefore, by the previous corollary,

$$v_2\left(d_K\right) \geq [K : \mathbb{Q}]$$

Then

$$d_{\mathbb{Q}(\alpha)} = d_K^{[\mathbb{Q}(\alpha):K]} \cdot N_{K/\mathbb{Q}}\left(d_{\mathbb{Q}(\alpha)/K}\right)$$

(see for example [4, III.§4]). Then

$$v_2\left(d_{\mathbb{Q}(\alpha)}\right) \geq [\mathbb{Q}(\alpha) : K] \cdot v_2(d_K)$$

However, $1, \alpha, \ldots, \alpha^{n-1}$ spans a sub lattice of the ring of integers of $\mathbb{Q}(\alpha)$ whose discriminant is d_n and hence

$$n \leq v_2\left(d_{\mathbb{Q}(\alpha)}\right) \leq v_2(d_n) = n - 1$$

and that's a glaring contradiction. $\square$

From this corollary, we can derive the following:

Corollary 2.8. *G acts primitively on* $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$.

Proof. Assume towards contradiction that the action is imprimitive. Then there is a strictly smaller non-trivial G set S and a surjective G-equivariant map

$$\pi : \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right) \to S$$

In particular, π is an epimorphism of G-sets, and hence by the famous equivalnece of categories between G-sets and Etale algebras over $\mathbb{Q}$, π

corresponds to a monomorphism of Etale algebras over $\mathbb{Q}$, i.e. a field embedding

$$\iota : K \hookrightarrow \mathbb{Q}(\alpha)$$

and since S is non-trivial and strictly smaller, K is a proper subextension of $\mathbb{Q}(\alpha)/\mathbb{Q}$, contradicting the previous corollary. $\square$

It immediately follows that G_n acts primitively on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$, and hence to completely seal the proof, it suffices to show the following lemma and a corollary:

Lemma 2.9. *A transitive permutation group that is generated by transpositions is the full symmetric group.*

Proof. Sketch inspired by a discussion with Itamar Israeli:

There is a seuqnece of transpositions that connects any two elements. Given a product of a pair of transpositions

$$(a\ b)(b\ c) = (a\ b\ c)$$

it can be shoretned to

$$(b\ c)(a\ b)(b\ c) = (a\ c)$$

□

Corollary 2.10. *A primitive permutation group that contains a transposition is the full symmetric group.*

Proof. Slightly inspired by Borys's thesis:

Let H be a primitive permutation group that contains a transposition τ. Let K be the normal closure of $\langle \tau \rangle$ inside H. Then the orbits of K form a system of blocks for H. Therefore, K is itself transitive and it is generated by transpositions and hence by the previouse lemma $K = S_n$ and hence $H = S_n$ as required. □

3 Background

Lemma 3.1. *Let $f \in \mathbb{Z}[x]$, and let L be its splitting field over $\mathbb{Q}$. Assume that over $\mathbb{F}_p$, the polynomial $\overline{f}$ decomposes as*

$$\overline{f}(x) = \prod_{i=1}^{k} \overline{f_i}(x)^{e_i}$$

where the $\overline{f_i}$ are distinct monic irreducible polynomials. Let $\mathfrak{P}$ be a prime of $\mathcal{O}_L$ lying over $(p) \subseteq \mathbb{Z}$. Then there is a partition, of the set

of roots of f into disjoint subsets $S_1, \ldots, S_k$ such that:

1. *for each $1 \leq i \leq k$, $|S_i| = e_i \deg\left(\overline{f_i}\right)$*

2. *The map*

$$\pi_{\mathfrak{P}} : \mathcal{O}_L \to \mathcal{O}_L/\mathfrak{P}$$

   reduces to a surjective e_i to 1 map from S_i onto the set of roots of $\overline{f_i}$ over $\mathbb{F}_p$.

3. *The decomposition group at $\mathfrak{P}$, $D(\mathfrak{P})$, preserves each S_i*

4. *The decomposition group at $\mathfrak{P}$ acts on S_i in a way that respects the partition into fibers of $\pi_{\mathfrak{P}}$.*

5. *The inertia group at $\mathfrak{P}$ acts trivially on the set of fibers of $\pi_{\mathfrak{P}}$.*

6. *The preimage of the frobenious automorphism $\phi : x \mapsto x^p$ under $\pi_{\mathfrak{P}}$ acts on S_i that respects the partition into fibers of $\pi_{\mathfrak{P}}$, and induces on the set of fibers a cycle of length $\deg\left(\overline{f_i}\right)$.*

Proof. By Hensel's lemma there are $f_1, \ldots, f_k \in \mathbb{Z}_p[x]$ such that for each $1 \leq i \leq k$, $\left(\overline{f_i}\right)^{e_i}$ is the image of f_i under the cannonical map

$\mathbb{Z}_p[x] \to \mathbb{F}_p[x]$, and

$$f(x) = \prod_{i=1}^{k} f_i(x)$$

over $\mathbb{Z}_p$. Set S_i to be the set of roots of f_i in L.

Fix $1 \leq i \leq k$ and let $\beta_1, \ldots, \beta_{\deg(\overline{f_i})}$ be the roots of $\overline{f_i}$ over $\mathbb{F}_p$. Then

$$\begin{aligned} &\prod_{\alpha \in S_i} (x - \pi_{\mathfrak{P}}(\alpha)) = \\ &\pi_{\mathfrak{P}}\left(\prod_{\alpha \in S_i} (x - \alpha)\right) = \\ &\pi_{\mathfrak{P}}(f_i(x)) = \\ &\overline{f_i}(x)^{e_i} = \\ &\prod_{j=1}^{\deg(\overline{f_i})} (x - \beta_j)^{e_i} \end{aligned} \tag{1}$$

and hence, since $\mathbb{F}_p^{\mathrm{alg}}[x]$ is a unique factorization domain, for each $1 \leq j \leq \deg(\overline{f_i})$, there are exactly e_i elements $\alpha \in S_i$ such that $\pi_{\mathfrak{P}}(\alpha) = \beta_j$. This proves (2) and in particular it implies (1).

Let $\sigma \in D(\mathfrak{P})$. Then σ extends to an automorphism $\widehat{\sigma}$ of the completion of L with respect the $\mathfrak{P}$-valuation $v_{\mathfrak{P}}$. We shall denote this completion by $\widehat{L}$. Since $\widehat{\sigma}$ fixes $\mathbb{Q}$, it also fixes its closure with

respec to $v_{\mathfrak{P}}$, a closure which is also known as $\mathbb{Q}_p$. In particular, the automorphism it induces on $\widehat{L}[x]$ fixes $\mathbb{Z}_p[x]$ and thus it fixes f_i and consequently so does the automorphism that σ it induces $L[x]$. Therefore,

$$\begin{aligned}
&\prod_{\alpha \in S_i} (x - \sigma(\alpha)) = \\
&\sigma\left(\prod_{\alpha \in S_i} (x - \alpha)\right) = \\
&\sigma(f_i) = \\
&f_i = \\
&\prod_{\alpha \in S_i} (x - \alpha)
\end{aligned}$$

and hence, since $L[x]$ is a unique factorization domain, it follows that $\sigma(S_i) = S_i$. Since the same is true for any i and for any $\sigma \in D(\mathfrak{P})$, it follows that $D(\mathfrak{P})$ preserves each S_i and.

It is well known that each element of the decomposition group induces an automorphism of the $L/\mathfrak{P}$, which in particular implies that it respects the partition of L, and in particular of S_i into fibres of $\pi_{\mathfrak{P}}$. The automorphism that any element of the inertia group, $I(\mathfrak{P})$,

induces on $L/\mathfrak{P}$ is the identity, by the mere definition of the inertia group, and hence the bijection that any such automorphism induces from the set of fibres to iteself is also the identity.

By a slight abuse of notation, use ϕ to denote some member of $D(\mathfrak{P})$ that induces the frobenious map ϕ on $L/\mathfrak{P}$ (it is a well known fact that any automorphism of $L/\mathfrak{P}$ is induces by some automorphism of L). Then as we have already shown, sicne it is a member of $D(\mathfrak{P})$ its action on S_i preserves the partition into fibres of $\pi_{\mathfrak{P}}$. Moreover, the action it induces on them is the action it induces on their images in $L/\mathfrak{P}$ which is indeed a cycle of order

$$[L/\mathfrak{P} : \mathbb{F}_p] = \deg\left(\overline{f_i}\right)$$

$\square$

Definition 3.2. *In the situation of lemma 3.1, we will call the each S_i a decomposition block of $\mathfrak{P}$ and we will call the intersection of each S_i with each fibre under $\pi_{\mathfrak{P}}$ an inertia block of $\mathfrak{P}$.*

4 Reduction to a 2-adic Question

For each $n \in \mathbb{Z}_{>1}$ denote

$$f_n(x) = x^{n-1} - \sum_{i=0}^{n-2} x^i$$

$$g_n(x) = (x-1) f_n(x) = x^n - 2x^{n-1} + 1$$

We will ommit for now the proof [or even any reference to that fact] that f_n is irreducible over $\mathbb{Q}$. Instead, we will reduce the question of whether its Galois group is the full symmetric group to a 2-adic question. Denote by L_n the splitting field of f_n over $\mathbb{Q}$ and by $G_n = \mathrm{Gal}(L_n/\mathbb{Q})$ its Galois group.

Lemma 4.1. *Let $\mathfrak{P} \subseteq \mathcal{O}_{L_n}$ be a prime at which f_n ramifies that lies over an odd prime $(p) \subseteq \mathbb{Z}$. Then the inertia group of f_n at $\mathfrak{P}$ is generated by a single transposition.*

Proof. It suffices to show that over $\mathbb{F}_p$, the polynomial f_n has exactly one double root and all other roots are simple. Indeed, assume that this is the case and denote by α the unique double root of f_n over $\mathbb{F}_p$. Then, any other root β of f_n over $\mathbb{F}_p$ lifts to a single root, and hence any

element of the inertia group must fix it. Therefore, the inertia group is embedded into the permutation group of the two lifts of α, and hence is either trivial or generated by a single transposition. However, since by assumption f_n ramifies at p, the inertia group is non-trivial, and hence it is indeed generated by the transposition of the two lifts of α.

We shall show then that f_n has exactly one double root over $\mathbb{F}_p$ and all other roots are simple. If α is a multiple root of f_n over $\mathbb{F}_p$, it is in particular a multiple root of g_n and hence

$$0 = g_n'(\alpha) = n\alpha^{n-1} - 2(n-1)\alpha^{n-2}$$

and hence, since $\alpha \neq 0$,

$$\alpha = \frac{2(n-1)}{n}$$

and it is in particular unique.

Since f_n is by assumption ramified at p, such α exists. We will show that the multiplicity of α is not greater than 2. It suffices to show that g_n has no root of multiplicity greater than 2, and since all of its roots are non-zero, its multiset of roots is obtained from the multiset of $h_n(x) = x^n - 2x + 1$ by inversion, and hence it suffices

to show that h_n has no root of multiplicity greater than 2 over $\mathbb{F}_p$. However, any root of multiplicity greater than 2 must also be a root of $h_n''(x) = n(n-1)x^{n-2}$, and hence must be 0, which is not a root of h_n. $\square$

Lemma 4.2. *Let $L/\mathbb{Q}$ be a Galois extension. Then*

$$G = \sum_{\mathfrak{P} \subseteq \mathcal{O}_L} I_{\mathfrak{P}}$$

where $I_{\mathfrak{P}}$ is the inertia group at the prime $\mathfrak{P}$.

Proof. The only unramified extension of $\mathbb{Q}$ is $\mathbb{Q}$ itself, and hence the fixed field of $\sum_{\mathfrak{P}} I_{\mathfrak{P}}$ is $\mathbb{Q}$. $\square$

Lemma 4.3. *A transitive permutation group generated by transpositions is the full symmetric group.*

Proof. Sketch inspired by Itamar Israeli: There is a seuqnece of transpositions that connects any two elements. Given a product of a pair of transpositions

$$(a\ b)(b\ c) = (a\ b\ c)$$

it can be shoretned to

$$(b\ c)\,(a\ b)\,(b\ c) = (a\ c)$$

$\square$

Corollary 4.4. *If n is odd, then $G_n = S_{n-1}$.*

Corollary 4.5. *Assume that n is even. To prove that $G_n = S_{n-1}$, it suffices to show that for any prime $\mathfrak{P} \subseteq \mathcal{O}_{L_n}$ above 2, the inertia group $I_{\mathfrak{P}}$ is generated by a transposition.*

We can also derive a slightly different reduction.

Lemma 4.6. *A primitive transitive permutation group G containing a transposition is the full symmetric group.*

Proof. Since it is transitive and primitive, any non-trivial normal subgroup of G is also transitive. Indeed the orbits of any normal subgroup form a partition of the set preserved by G. Therefore, the normal subgroup generated by the transposition whose existence is assumed is transitive, and hence must be the full symmetric group, and it is generated by all the conjugates of that transposition, which are all transpositions in themselves. $\square$

Lemma 4.7. *If $n > 2$ then d_n is not a power of 2.*

Proof. By [2],

$$d_n = \pm \frac{n^n - 2^n (n-1)^{n-1}}{(n-2)^2}$$

If n is odd, then $n^n - 2^n (n-1)^{n-1}$ is odd as well, and hence d_n is not a power of 2. Assume that n is even, then by Lemma 5.3,

$$v_2(d_n) = n - 2$$

□

Corollary 4.8. *To prove that $G_n = S_{n-1}$, it suffices to show that G_n is primitive and that the discriminant of L_d over $\mathbb{Q}$ is not a power of 2 (for which is sufficient to show that d_n is not a power of 2).*

Lemma 4.9. *A permutation group G that contains an n-cycle and a transposition is symmetric.*

Proof. I guess we could use a version of Itamar's proof here as well □

5 Random Facts

5.1 The Polynomial is irreducible

Lemma 5.1. *f_n is irreducible over $\mathbb{Q}$.*

Sketch: Assume $n > 2$ (otherwise it is trivial). The winding number of g_n as x traverses the circle of radius $1+\epsilon$ is $n-1$ for sufficiently small $\epsilon > 0$. Indeed, for such ϵ,

$$|2x^{n-1}| > |x^n| + 1 \geq |x^n + 1|$$

since

$$(1+\epsilon)^n - 2(1+\epsilon)^{n-1} + 1 < 0$$

since

$$g'(1) = n - 2(n-1) = 2 - n < 0$$

Therefore, it has $n-1$ roots of absolute value at most 1. One of these roots is 1, and hence f_n has $n-2$ roots of absolute value at most 1.

Moreover, the only root of absolute value exactly 1 is 1 itself. Indeed, it suffices to show that the only root of $\hat{g_n}(x) := g_n(1/x)\, x^n =$

$x^n - 2x + 1$ of absolute value 1 is 1 itself (since the roots of $\hat{g_n}$ are the inverses of the roots of g_n). However, any algebraic number α of absolute value 1 is a root of unity, and hence if $g_n(\alpha) = 0$ then $1 - 2\alpha$ is a root of unity as well, and in particular, by the inverse triangle inequality,

$$1 = |2\alpha - 1| \geq 2 - 1 = 1$$

and hence $\alpha = 1$.

Therefore, f_n has $n - 2$ roots of absolute value strictly less than 1. In other words, only one of its roots has absolute value greater than or equal to 1 (in fact greater). If we assume towards contradiction that f_n is reducible over $\mathbb{Q}$, then since it is monic, it is reducible over $\mathbb{Z}$ as well, and hence one of its factors is a monic polynomial with integer coefficients whose roots are all of absolute value strictly less than 1 which is a contradiction (because then for example the norm, which is the free coefficient up to a sign, is less than 1 in absolute value). $\square$

5.2 The Action of the Inertia Group on the Roots of f_n

Lemma 5.2. *Let $\mathfrak{P}$ be a prime of $\mathcal{O}_{L_n}$ lying above $(2) \subseteq \mathbb{Z}$. Then the inertia group $I_{\mathfrak{P}}$ acts transitively on each inertia block of $\mathfrak{P}$.*

Proof. Tweak the proof to apply to the tamely ramified case as well

Let B be an inertia block of $\mathfrak{P}$. It suffices to show that all elements of B are conjugate over $\mathbb{Q}_2^{\mathrm{un}}$, the maximal unramified extension of $\mathbb{Q}_2$. To show that in turn, it suffices to show that the degree of one of these elements over $\mathbb{Q}_2^{\mathrm{un}}$ is at least the size of B. Indeed, assume that $\alpha \in B$ and assume that the degree of α over $\mathbb{Q}_2^{\mathrm{un}}$ is at least $e = |B|$. Assume that β is a conjugate of α over $\mathbb{Q}_2^{\mathrm{un}}$. Then there is an automorphism σ of $\mathbb{Q}_2^{\mathrm{alg}}$ over $\mathbb{Q}_2^{\mathrm{un}}$ such that $\sigma(\alpha) = \beta$. In particular, it reduces to a member of the inertia subgroup of $\mathrm{Gal}(L_n/\mathbb{Q})$ at $\mathfrak{P}$ that maps α to β and hence α and β are equivalent modulo $\mathfrak{P}$ and hence $\beta \in B$. Therefore, for the number of conjugates of α over $\mathbb{Q}_2^{\mathrm{un}}$ to reach e, all elements of B must be conjugate to α over $\mathbb{Q}_2^{\mathrm{un}}$.

It remains to show that there is a root $\alpha \in B$ whose degree over $\mathbb{Q}_2^{\mathrm{un}}$ is at least e or equivalently that there is $\beta \in \mathbb{Q}_2^{\mathrm{un}}(\alpha)$ whose 2-adic

valuation is in the interval $(0, 1/e]$. Indeed, let $k = n/2^e$ and let β be the unique primitive k-th root of unity whose image under $\pi_{\mathfrak{P}}$ is the image of B under $\pi_{\mathfrak{P}}$. Then the free coefficient of $g_n(x+\beta)$ is

$$g_n(\beta) = \beta^n - 2\beta^{n-1} + 1 = 2\left(1 - \beta^{n-1}\right)$$

whose 2-adic valuation is exactly 1. Therefore, since $g_n(x+\beta)$ has exactly 2^e roots whose $\mathfrak{P}$-valuation is strictly positive, by Newton's polygon argument, there is a root α of $g_n(x+\beta)$ whose $\mathfrak{P}$-valuation is not more than $1/2^e$. $\square$

5.3 The Discriminant of f_n

Lemma 5.3. *Assume that $n \geq 3$ is even then*

$$v_2(d_n) = n - 2$$

Proof. By [2]

$$d_n = \frac{(-1)^{\binom{n}{2}} \left(n^n - 2^n (n-1)^{n-1}\right)}{(n-2)^2} = (-1)^{\binom{n}{2}} 2^n \cdot \frac{\left(\frac{n}{2}\right)^n - (n-1)^{n-1}}{(n-2)^2}$$

and hence if we denote $m = v_2(n-2)$ we get that

$$v_2(d_n) = n - 2m + v_2\left(\left(\frac{n}{2}\right)^n - (n-1)^{n-1}\right)$$

Therefore, it suffices to show that

$$v_2\left(\left(\frac{n}{2}\right)^n - (n-1)^{n-1}\right) = 2m - n + n - 2 = 2m - 2$$

Assume that $m \geq 2$ Indeed,

$$n - 2 \equiv \epsilon 2^m \mod 2^{2m-1}$$

for some odd ϵ, and hence

$$n \equiv \epsilon 2^m + 2 \mod 2^{2m-1}$$

so

$$
\begin{aligned}
&\left(\frac{n}{2}\right)^n - (n-1)^{n-1} \equiv \\
&\left(\epsilon 2^{m-1}+1\right)^n - \left(\epsilon 2^m+1\right)^{n-1} \equiv \\
&\binom{n}{2}\epsilon^2 2^{2m-2} + n\epsilon 2^{m-1} + 1 - (n-1)\,\epsilon 2^m - 1 \equiv \backslash 3m-3 \geq 2m-1 \\
&\binom{n}{2}\epsilon^2 2^{2m-2} + \epsilon 2^m\left(\frac{n}{2} - (n-1)\right) \equiv \\
&\binom{n}{2}\epsilon^2 2^{2m-2} + \epsilon 2^m\left(\epsilon 2^{m-1} + 1 - \epsilon 2^m - 1\right) \equiv \\
&\binom{n}{2}\epsilon^2 2^{2m-2} + \epsilon 2^m\left(-\epsilon 2^{m-1}\right) \equiv \\
&\binom{n}{2}\epsilon^2 2^{2m-2} - \epsilon^2 2^{2m-1} \equiv \\
&\left(\binom{n}{2} - 2\right)\epsilon^2 2^{2m-2} \mod 2^{2m-1}
\end{aligned}
$$

and in particular its v_2 is $2m-2$ as required.

Assume finally now that $m = 1$, then

$$v_2\left(\frac{n}{2}\right) \geq 1$$

and hence

$$v_2\left(\left(\frac{n}{2}\right)^n\right) \geq n > 0 = v_2\left((n-1)^{n-1}\right)$$

and hence

$$v_2\left(\left(\frac{n}{2}\right)^n - (n-1)^{n-1}\right) = 0 = 2m - 2$$

Assume that $m = 2$, then

$$v_2\left(n^n\right) \geq 2n > v_2\left(2^n (n-1)^{n-1}\right)$$

and hence

$$v_2\left(n^n - 2^n (n-1)^{n-1}\right) = n$$

and hence

$$v_2\left(d_n\right) = n - 2$$

□

Corollary 5.4. *There is an odd prime that divides d_n for any $n \geq 4$.*

Proof. Sketch: The maximal odd divisor of d_n is

$$\frac{n^n - 2^n (n-1)^{n-1}}{2^{v_2\left(n^n - 2^n (n-1)^{n-1}\right)} (n-2)^2} =$$
$$\frac{n^n - 2^n (n-1)^{n-1}}{2^{n-2} (n-2)^2} =$$
$$\frac{\left(\frac{n}{2}\right)^n - (n-1)^{n-1}}{\left(\frac{n-2}{2}\right)^2}$$

and the numerator clearly clearly grows faster than the denominator.

□

5.4 Ramification at 2

Lemma 5.5. *Let $n \geq 4$ be even. Then L_n ramifies at 2.*

Proof. Let v be a valuation extending v_2 to L_n and let $\widehat{L}$ be the completion of L_n with respect to v. It suffices to show that □

6 Random Calculations

6.1 The case $(n = 3)$

This case is the just the classic Golden Ratio calculation. Explicitly,

$$f_3(x) = x^2 - x - 1 = 0$$

its roots are given by

$$\alpha = \frac{1+\sqrt{5}}{2}, \quad \beta = \frac{1-\sqrt{5}}{2}$$

The difference between them is $\sqrt{5}$ which indeed generates the unique prime ideal above 5 in $\mathcal{O}_{\mathbb{Q}(\sqrt{5})}$ and thus the unique element of $G(f_3)$ is ramified above 5 as expected. Explicitly, both solutions are mapped to $3 \in \mathbb{F}_5$ and hence the unique of $G(f_3)$ acts trivially on the residue field.

In a more modern (but maybe in this case redundant) language, we could note that α generates $\mathcal{O}_{\mathbb{Q}_5(\sqrt{5})}$ over $\mathcal{O}_{\mathbb{Q}_5}$. Indeed, following the procedure presented in Serre's proof of [4, III §6 Proposition 12], we can note that

$$v_5(\alpha + 2) = v_5\left(\frac{5+\sqrt{5}}{2}\right) = 1$$

and hence

$$1, \alpha, \alpha + 2, \alpha^2 + 2\alpha$$

generate $\mathcal{O}_{\mathbb{Q}_5(\sqrt{5})}$ over $\mathcal{O}_{\mathbb{Q}_5}$ as a module [maybe I need to show it explicitly at some point]. Hence the fact that the unique element of $G(f_3)$ maps α to β which is in the same $\left(\sqrt{5}\right)$ coset, implies that it belongs to the ramification group.

6.2 The case $(n = 5)$

First off, a direct calculation shows that

$$d_5 = \pm 563$$

in particular, it ramifies only above 563. Moreover, in $\mathbb{F}_{563}$, the unique double root is

$$\alpha := \frac{2 \cdot 4}{5} \equiv 8 * 338 \equiv 452 \mod 563$$

6.3 The Case $n = 10$

Let L be the splitting field of f_{10} over $\mathbb{Q}_2$.

Lemma 6.1. *L is ramified over* (2).

Proof. It suffices to show that $\mathcal{O}_L$ contains an element of fractional valuation. Note that over $\mathbb{F}_2$,

$$\overline{g_{10}}(x) = x^{10} + 1 = \left(x^5 + 1\right)^2 = (x+1)^2 \left(x^4 + x^3 + x^2 + x + 1\right)^2$$

Therefore, the roots of g_{10} over $\mathbb{Q}_2$ are given by two lifts of 1 and two

lifts of each root of $x^4+x^3+x^2+x+1$ over $\mathbb{F}_{2^4}$. Choose $\overline{\beta}$ to be a root of $x^4+x^3+x^2+x+1$ in $\mathbb{F}_{2^4}$. Let $\beta \in \mathcal{O}_L$ be a lift of $\overline{\beta}$ (not necessarily a root of g_{10}). Consider the polynomial,

$$f_{10}\left(x+\beta\right)$$

Its roots are given by $\alpha_i - \beta$ where α_i are the roots of f_{10}. Therefore, if we fix a prime $\mathfrak{P} \subseteq \mathcal{O}_L$ lying over (2), by Lemma 3.1, $f_{10}\left(x+\beta\right)$ has exactly two roots in $\mathfrak{P}$. $\square$

We will show that G_{10} is primitive. Let $\alpha_1, \ldots, \alpha_9$ be the roots of f_{10}. Then in $\mathbb{F}_2$,

$$\begin{aligned}
&\overline{(x-1)\prod_{i=1}^{9}\left(x-\alpha_i\right)} = \\
&\overline{g_{10}}\left(x\right) = \\
&x^{10}-2x^9+1 = \\
&x^{10}+1 = \\
&\left(x^5+1\right)^2 = \\
&\left(x+1\right)^2\left(x^4+x^3+x^2+x+1\right)^2
\end{aligned}$$

6.4 The case $n = 2p$ for p an odd prime

We will show that G_n is primitive. Assume towards contradiction that there is a partition of the $n-1$ roots of f_n into blocks respceted by G_n. Since G_n acts transitively on the roots, all blocks must be of the same size, say d, and denote the blocks by $B_1, \ldots, B_k$ where $k = \frac{n-1}{d}$.

Note now that over $\mathbb{F}_2$, g_n factors as

$$g_n = x^n + 1 = (x^p + 1)^2 = (x+1)^2 \left(\sum_{i=0}^{p-1} x^i \right)^2$$

and hence

$$\overline{f_n} = \overline{\frac{g_n}{x-1}} = (x+1) \left(\sum_{i=0}^{p-1} x^i \right)^2$$

Fix a prime Q of $\mathcal{O}_{L_n}$ lying over (2). Let α be the unique root of f_n that projects to 1 and assume WLOG that $\alpha \in B_1$. Since it forms its own decomposition block, the decomposition group at Q stabilizes α. In particular, it stabilizes B_1. Therefore, since by lemma [will be added later], the inertia group at Q acts transitively on each inertia block, the rest of B_1 is a union of $\frac{d-1}{2}$ inertia blocks. Therefore, since the frobenious automorphism at Q acts on the set of inertia blocks as

a cycle of length $\mathrm{ord}_p(2)$, it follows that $\mathrm{ord}_p(2) \mid \frac{d-1}{2}$.

Now, fix $1 < i \leq k$. Since $\#B_i = d$ is odd, B_i cannot be a union of inertia blocks, and hence it must contain at least one root, α such that there is at least one automorphism σ that maps α to a root in B_j for some $j \neq i$. However, since G respects the block structure, σ must map the entire block B_i to B_j. Therefore, for each root $\beta \in B_i$, the unique other root that shares its inertia block must also be in B_j.

Let $\phi \in D(\mathfrak{P})$ be the frobenious. Then one of two options must hold:

1. $\phi(B_i) = B_i$. But this implies that $\mathrm{ord}_p(2) \mid d$ which is impossible since then $\mathrm{ord}_p(2) \mid \gcd\left(d, \frac{d-1}{2}\right) = 1$.

2. There is $\alpha \in B_i$ such that $\phi(\alpha) \in B_j$ for some $j \neq i$.

and denote by $I \leq G_n$ the inertia group corresponding to Q. Then the map π_Q reduces to a surjective 2 to 1 map from the roots of g_n to the $\mu_{p-1} \subseteq \overline{\mathbb{F}_2}$ and the action of I stabilizes the fibers of this map.

In particular, the set of roots of g_n includes exactly two lifts of 1 which the action of I either fixes or swaps. However, $1 \in \mathbb{Q}$ must be one of these two lifts, and hence I must fix both of them. Denote the

second lift of 1 by α and denote by B_1 the block containing α. Then since G_n respects the block structure, I must stabilize B_1.

By lemma [will be added later], the action of I on each inertia block is transitive. Therefore,

6.5 The case $n = 2^d$

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